🥽 CSE 165 - 3D User Interactions (Spring 2026)

Project 1 - Selection, Manipulation and Travel 👉

Deadline: 09:00am, Friday, April 24, 2026

This project has three parts: A design exploration, an implementation, and a heuristic evaluation. The project also includes the opportunity for extra credit.

You are expected to do this project in teams of two.

This is a graded assignment. You can obtain 100 points total, plus up to 10 points of extra credit.

For this project you should use Unity 3D (https://unity3d.com/) for the implementation.

Design and implement a VR application allowing the user to be immersed in a life-size 3D model of an empty tent of an emergency medical site. The user can furnish the space with different medical devices and equipment, which serves as guidance to the novice user on how to arrange the emergency medical site.

You will need to submit by the assignment due date:

- A Project Design Document that has your design explorations and heuristic evaluation. A template can be found from here.
- A Github link of your code. Please make sure your repository is publicly accessible. If this is the first time you are using Github, feel free to claim the free Github Education package with your university email address.
- A video showing all required features.

Design Exploration (10 points)

Brainstorm different interaction techniques that the user could use in this project. Develop a storyboard for each interaction technique. An example of the design of a selection technique can be found in Figure 1. You can sketch on paper or use tools like Figma (https://www.figma.com/).

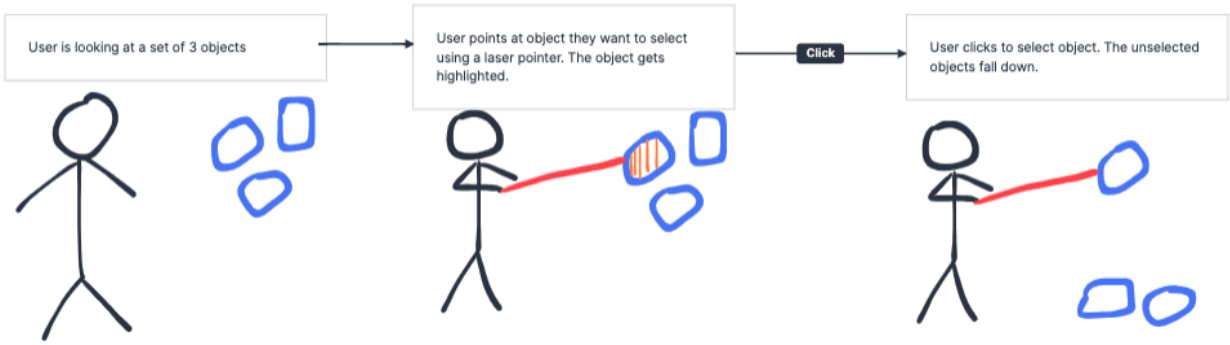

Figure 1: Example storyboard of designing a selection technique.

You must submit at least 3 storyboards for selection and manipulation techniques, and 3 storyboards for travel techniques. Before brainstorming your design, we recommend referring back to our lecture slides. Add these images to your Project Design Document.

Design Exploration Rubric

- There is at least 1 design idea in the submission (1 point)
- There are at least 3 different Selection & Manipulation storyboards (0.5 points each) -- (1.5 points total)
- There are at least 3 different Travel storyboards (0.5 points each) -- (1.5 points total)
- Each storyboard is relevant (0.5 points each) and viable (0.5 points each) -- (6 points total)

Implementation (70 points + 10 extra credit points)

In this project, the only permitted interaction devices are the Meta Quest 2 headset, along with two Quest controllers. To reduce the cognitive load on the user, we have decided to restrict the usage of buttons on the controller in our design. Only the trigger button and the grip button are allowed to be used for the interaction with objects (selection and manipulation) in this project (see Figure 2). While implementing your design, you are allowed to use the off-the-shelf packages such as Interactors and Interactables. Only OpenXR SDK and Unity are allowed in this project. If this is the first time you are using OpenXR, please first go through the documentation in [1].

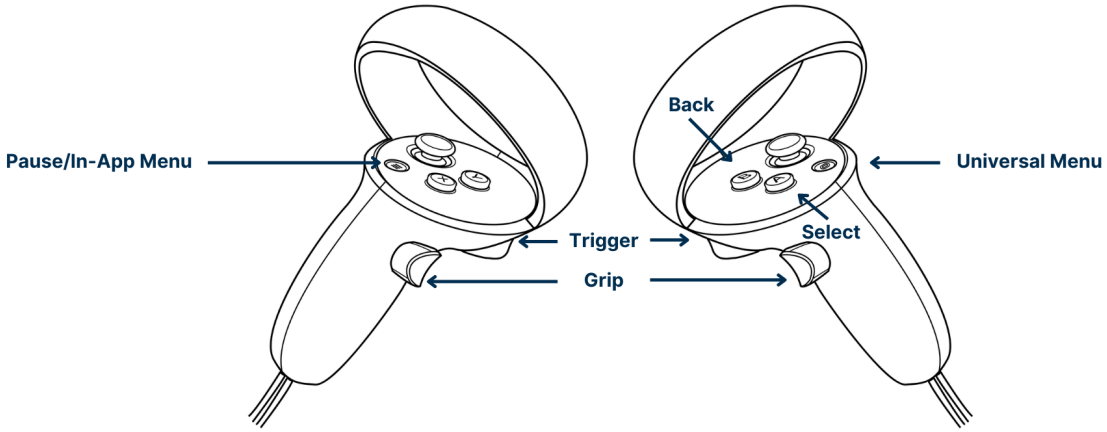

Figure 2: Button mapping of the Meta Quest 2 controller

Imagine that a user has scanned an empty tent in a rural area where they need to set up a temporary emergency medical site. The medical expert, wearing the VR headset, will be tele-presented to the scanned environment and tasked with designing the layout and arrangement of the medical devices and equipment.

To simplify the setup, we have provided the 3D model of the emergency medical site. You can import the lifesize model using this unity package: HW1.unitypackage, containing the tent, as well as various medical devices. Figure 3 shows the first-person view of the 3D model.

Figure 3: First-person view of the medical

Achieving the basic functionalities requires implementing following key features:

1. **Initializing**: Place the user in the center of the tent *(5 points)*.
2. **Spawning**: Allow the users to spawn two different items of your choices *(3 + 2 = 5 points)*, and provide a way for the user to modify the placement and orientation *(5 points for placement and 5 points for orientation)*. Once spawned, the objects should have physics properties attached to them and react to gravity and collisions *(5 points)*.
3. **Selection and Manipulation**: Allow the users to select already placed items in the space and highlight the corresponding selections *(5 points for selection, 5 points for highlight)*. We expect two different interaction schemes for selection (5 points for the second selection method). Allow manipulation of the highlighted objects. After an item is selected, design a way to allow the users to move, rotate and scale the item *(3 + 3 + 4 = 10 points)*. Only the trigger button and the grip button are allowed to be used. *(-5 out of 10 if other buttons are used for manipulation)*
4. **Travel**: The user should be able to move around the space allowing free movement from one side to the other and control over orientation *(5 points movement, 5 points orientation)*
5. **Indicator**: It is not necessary to use a ray, but you should include an indicator/pointer for each corresponding selection and travel method *(5 points for selections, 5 points for travel)*.

Implementation Extra Credit (10 points)

There are options for extra credit. You can get a maximum of 10 points for extra credit.

1. **Improve the Spawning Method**: Design a way to allow the user to spawn a large variety of different items in the space. It can be a menu, gesture or anything else *(5 points)*
2. **Two-hand interaction**: Design a two-hand interaction system that allows the user to perform complex selection and manipulation tasks, such as selecting heavy-occluded objects behind other objects or performing non-rigid transformations *(5 points)*.
    - 3 points if only selection or only manipulation.
3. **Group Selection and Manipulation**: Design a way for the user to select more than one item in the scene and manipulate them all at once. We expect to see translation, orientation, scaling and duplication for the manipulation *(5 points)*.
    - 3 points selection, 2 points manipulation
4. **Hand Gesture for Selection and Manipulation**: Sometimes interactions with hands directly feel more natural than using a controller. Utilize the hand tracking capability

of the Meta Quest 2 and design a way for the user to select and manipulate items in the scene *(5 points)*.

Heuristic Evaluation (20 points)

Once you have finished implementing your project, you should conduct a heuristic evaluation of the system using the template in your Project Design Document.

There are a total of 10 heuristics (worth 2 points each) for which you need to input:

1. **Violation + Rationale**: Does your system violate the heuristic? Why do you think so? (1 point each)
2. **Severity**: How much does this violation impact the user experience? (0.25 points each)
3. **Recommendation**: How can you fix this issue? If any of your techniques from the design exploration work better, mention it. (0.75 points each)

References

[1] Unity OpenXR: Meta: https://docs.unity3d.com/Packages/com.unity.xr.meta-openxr@1.0/manual/index.html

[2] Github Education: https://github.com/education